Subdivision-Respecting Redistricting: County-Sticky Edge Weights

B.10 — Algorithm Track

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Abstract

Most state constitutions require congressional maps to "preserve political subdivisions" — counties and municipalities — as a secondary criterion after equal population. Existing algorithmic approaches either ignore this requirement or impose hard constraints that degrade compactness. We introduce *county-sticky* edge weighting: edges between census tracts within the same county receive a multiplier $\alpha_c \geq 1$, making intra-county cuts more expensive without prohibiting them. On all 50 states, $\alpha_c = 3.0$ reduces county splits by 34% relative to the baseline (B.2) while maintaining 97% of the compactness improvement. This soft-constraint approach is the weight mode exposed as `-weights-override county` in the `redist` CLI.

1 Introduction

State constitutional redistricting criteria typically list criteria in priority order: (1) equal population, (2) VRA compliance, (3) contiguity, (4) compactness, (5) preservation of political subdivisions. Items (1)–(4) are addressed by the baseline algorithm (B.2). Item (5) — county preservation — requires explicit treatment.

Two naive approaches fail: *Hard constraints* (forbidden edges across county boundaries) make large counties unpartitionable when $k >$ county count, breaking contiguity. *No treatment* (ignore counties) produces maps that split counties unnecessarily, violating state constitutional requirements.

We propose a middle path: *county-sticky weights*.

2 County-Sticky Weighting

2.1 Formulation

Let $c(v)$ denote the county of tract v. For each edge $(u, v) \in E$, define the county-sticky weight:

$$w'_E(u, v) = w_E(u, v) \cdot \begin{cases} \alpha_c & \text{if } c(u) = c(v) \\ 1 & \text{otherwise} \end{cases}$$

where $w_E(u, v)$ is the shared boundary length (B.2 baseline) and $\alpha_c \geq 1$ is the county stickiness parameter.

Higher α_c penalises intra-county cuts more, preserving county integrity at the cost of compactness.

2.2 Parameter selection

We sweep $\alpha_c \in \{1.5, 2.0, 3.0, 5.0, 10.0\}$ on all 50 states. The Pareto frontier of (county splits, compactness) shows a sharp elbow at $\alpha_c = 3.0$: further increases reduce county splits by $< 5\%$ while compactness falls by $> 8\%$.

3 Results

Table 1: County preservation vs compactness ($\alpha_c = 3.0$, 2020 Census)

Metric	Baseline (B.2)	County-Sticky	Change
Mean Polsby–Popper	0.361	0.350	−3.0%
County splits	487	323	−34%
Multi-county districts	312	198	−37%
Max deviation from ideal pop.	0.41%	0.44%	+0.03pp

County preservation improves substantially while compactness falls only 3% — well within the margin of improvement over enacted maps (+22% vs B.2 baseline).

3.1 State-level highlights

Iowa (k=4, 99 counties): county splits drop from 12 to 3. The four-district map nearly respects the traditional four-region Iowa geography (northwest, northeast, southwest, southeast).

Texas (k=38, 254 counties): county splits drop from 89 to 61. The 15 largest counties still require splitting (≥ 4 districts each), but medium counties are now largely preserved.

4 Connection to the Districting Integrity Act

The DIA's algorithm specification uses county-sticky weights ($\alpha_c = 3.0$) as a default, satisfying state constitutional subdivision-preservation requirements without hard constraints. The parameter α_c is published in the TIGER adjacency graph metadata, making the weighting fully reproducible.

5 Conclusion

County-sticky edge weights with $\alpha_c = 3.0$ reduce county splits by 34% at a compactness cost of 3%, satisfying state constitutional subdivision-preservation criteria without sacrificing the core compactness gains of edge-weighted bisection. This approach is implemented as `-weights-override county` in the `redist` CLI.

References